Status	Finished
Started	Wednesday, 21 October 2026, 9:18 PM
Completed	Wednesday, 21 October 2026, 9:24 PM
Duration	6 mins 11 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(1) ✓
- b. O(n log n)
- c. O👎
- d. O(log n)

Question **2**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the search algorithm finishes
- b. When the list is sorted
- c. When the element is found in the list ✓
- d. When the list contains duplicate elements

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

- a. There should be direct access to the middle element in any sublist
- b. The list must be sorted
- c. Number values should only be present

  Number values should only be present

- d. There must be a mechanism to delete and/or insert elements in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. n/2
- c. log n
- d. n-1

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94
- b. 90 and 99 ✓
- c. 89 and 99
- d. 89 and 94

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is somewhere in the middle of the array ✓
- c. When the item is not the array at all
- d. Item is the last element in the array or item is not there at all

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Must use a sorted array

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. There must be a mechanism to access middle element directly

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the right sublist ✓

b. In the left sublist

c. In the middle of the list

d. At the beginning of the list

Question **9**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned

b. The search is considered unsuccessful ✓

c. The last element is returned

d. An error is raised

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓

b. 1

c. log n

d. n/2

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By dividing the list length by two ✓
- b. By starting from the first element
- c. By comparing each element sequentially
- d. By using a hash function

Question **12**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O(log n)
- b. O(n log n)
- c. O👎 ✓
- d. O(1)

Question **13**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Hash search
- c. Interpolation search
- d. Linear search ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search

b. Binary search

c. Linear search ✓

d. Both (1) & (3)

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search divides the list into halves.

c. Linear search can be applied to both sorted and unsorted lists. ✓

d. Linear search is more efficient than binary search on large lists.